120 Days of Quantitative Finance

Month 4: Stochastic Calculus

Week 13: Brownian Motion

Day 89: Ito's Lemma

1 Introduction

In classical calculus, we use the chain rule to differentiate functions of variables.

However, Brownian Motion is not differentiable, so classical calculus does not apply.

Ito's Lemma provides the correct framework for differentiating functions of stochastic processes.

2 Setup

Let X_t follow a stochastic process:

$$dX_t = \mu dt + \sigma dW_t$$

Let $f(X_t, t)$ be a function of X_t and time.

3 Ito's Lemma

Then:

$$df = \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial x}dX_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}\sigma^2 dt$$

Substituting dX_t:

$$df = \left(\frac{\partial f}{\partial t} + \mu \frac{\partial f}{\partial x} + \frac{1}{2}\sigma^2 \frac{\partial^2 f}{\partial x^2}\right) dt + \sigma \frac{\partial f}{\partial x} dW_t$$

4 Key Insight

Compared to classical calculus:

$$df = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dx$$

Ito's Lemma introduces an additional term:

$$\frac{1}{2}\sigma^2 \frac{\partial^2 f}{\partial x^2} dt$$

This arises because:

$$(dW_t)^2 = dt$$

5 Example

Let:

$$f(W_t) = W_t^2$$

Using Ito's Lemma:

$$df = 2W_t dW_t + dt$$

This shows how stochastic calculus differs fundamentally from classical calculus.

6 Application in Finance

Ito's Lemma is used to:

- derive Black-Scholes equation
- transform price processes

- analyze stochastic differential equations

7 Conclusion

Ito's Lemma is the stochastic equivalent of the chain rule.

It is one of the most powerful tools in quantitative finance and is essential for working with continuous-time models.